

क्षणिक Kshanik

"क्षणिक" — transient, momentary · Physics-Informed Yield Surrogate for a Non-Isothermal Plug-Flow Reactor

A machine learning surrogate for reactor yield prediction, built by fitting the governing reaction kinetics and energy balance directly, then applying **Gaussian Process** and **CatBoost** models to correct only the residual error the mechanistic fit leaves behind.

Cross-validated RMSE 8.79 on 150 training observations

Team Transience · Fugacity 2026 ML Hackathon, IIT Kharagpur

· Preenithi Varshana · Aditya Kalagatoori · Kritika Pandey · 150 training rows, 50 held-out test rows

train_dataset.csv → mechanistic_pfr.py → warped GPR / CatBoost → overall_yield

25% ROWS AT EXACT ZERO YIELD	5 → 13 RAW INPUTS → PHYSICS FEATURES	8.79 BEST CV RMSE (WARPED GPR)	2 MODELS: SMALL-DATA & LARGE-DATA
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LIMITATION OF DIRECT REGRESSION

Black-Box Models Cannot Isolate the Yield Collapse

A regressor fit directly on `flow_rate`, `concentration`, `inlet_T`, `length`, and `jacket_T` must infer, from 150 rows alone, that yield falls to exactly **0%** whenever residence time and temperature drive the side reaction (B→C) past the desired one (A→B). No single raw feature correlates strongly with yield — the highest is $|r| = \mathbf{0.64}$ — because the governing variable is a non-linear combination the model has no basis to construct from data this limited.

TRANSCIENCE APPROACH

A Mechanistic Prior, Corrected by a Gaussian Process

We fit a non-isothermal PFR model directly — series kinetics **A→B→C** under Arrhenius rate laws, coupled to a jacket energy balance — using nonlinear least squares on the training set. Its output serves as the **prior mean** for a Gaussian Process, which is left to learn only the mechanistic model's residual error rather than the full yield surface.

MODEL PERFORMANCE — 5×10 REPEATED K-FOLD CV

MECHANISTIC MODEL ALONE

12.17

RMSE · physics only, no ML

ADDITIVE-RESIDUAL GPR

10.14

RMSE · mechanistic mean + plain GP

WARPED-LOGIT GPR

8.79

RMSE · selected small-data model

CATBOOST + SYNTHETIC

11.35

RMSE · large-data model

HOW IT WORKS — 5-STAGE MODELING PIPELINE

1

FIT ODE

Arrhenius kinetics and energy balance, fit by least squares on the training set

2

ENGINEER FEATURES

Damköhler numbers, selectivity ratio, residence time, ΔT derived from the fit

3

WARP TARGET

Logit transform of yield/100, respecting the [0, 100] boundary

4

LEARN RESIDUAL

Matérn-kernel Gaussian Process fits the mechanistic model's error

5

PREDICT

Inverse transform back to overall_yield for the test set

MECHANISTIC FOUNDATION

Non-Isothermal PFR Model

Series reaction $A \rightarrow B \rightarrow C$ under Arrhenius kinetics $k_1(T)$, $k_2(T)$, coupled to a jacket energy balance with explicit reaction-heat terms (Fogler, *Elements of Chemical Reaction Engineering*, Ch. 8). Fit as a grey-box model via multi-start nonlinear least squares, not assumed from literature constants.

Train RMSE: **12.17** ·
 $E_{a2} > E_{a1}$, consistent with side-reaction dominance at high temperature

SMALL-DATA MODEL

Warped Gaussian Process

The GP is trained in $\text{logit}(\text{yield}/100)$ space rather than on the raw residual. This is the statistically appropriate likelihood for a target bounded on $[0, 100]$ with substantial mass at the boundary, consistent with the censored-regression and warped-GP literature.

CV RMSE: **8.79** $\pm$ 3.87
· outperforms both additive-residual and classifier-gated variants

LARGE-DATA MODEL

CatBoost with Synthetic Augmentation

5,000 synthetic observations, sampled across the operating envelope and simulated through the fitted ODE, are combined with the real data at a $0.15\times$ training weight. A classifier stage first separates the collapsed-yield regime before a regressor is applied, a strategy intended to scale to a larger production dataset.

CV RMSE: **11.35** ·
versus 15.24 on real data alone, no augmentation